

10 MAINTENANCE AND INSPECTION

10.1 Summary

This chapter describes the maintenance and inspection procedures in order to maintain the weighing machine in good condition.

**WARNING**

- **For maintenance and inspections, unless instructed, the operator must turn OFF and lock the main power switch, and keep the key in his possession during the work.**
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<Contents>

- Daily inspection and periodical inspection
- Adjustment of components

<Purpose>

- To understand the inspection and adjustment procedures for the components in order to maintain the weighing machine in good condition and prevent loss of production.

<Intended reader>

- Operators
- Maintenance engineers

10.2 Daily Inspection

This section describes the daily inspection items.

10.2.1 Pre-start Inspection

Perform the following pre-start inspection before production.

WARNING

- If any part of the weighing machine is not securely installed, it may fall due to vibration during the production and may damage the weighing machine or injure personnel.

NOTE

- The installation methods for each part are explained in "12.4 Installation Procedure".

Table 10-1

Inspection Item	Inspection detail
Weighing machine and vicinity of weighing machine	Make sure that tools and any other irrelevant objects are not placed on top of or in the vicinity of the weighing machine.
Installing dispersion table	Make sure that the dispersion table is securely installed with no play.
Interference between dispersion table and radial trough	Make sure that the dispersion table does not make contact with the radial troughs (Refer to "10.2.1.1 Dispersion table").
Interference between radial troughs	Make sure the radial troughs are not touching each other (Refer to "10.2.1.2 Radial trough").
Installing hoppers	Make sure that the roller of each hopper securely fits into the U-shape groove of the switching lever.
Installing collection chute	Make sure that the collection chute is securely installed.
Installing timing hopper	Make sure that the timing hopper and the drive unit are securely connected with the drive lever.
Confirming the dew-point checker of the air dryer (WP specification)	Confirm performance of the air dryer by checking the grain color of the dew point checker (Refer to "10.2.1.3 Confirming the dew-point checker of the air dryer (WP specification)").

10.2.1.1 Dispersion table

For interference between the dispersion table and radial trough, check the circled portion shown in the figure below.

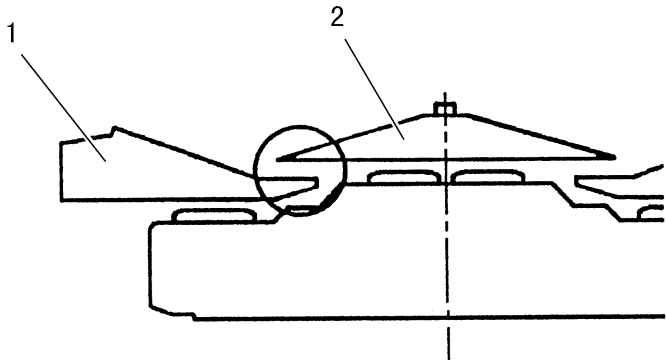


Fig. 10-1

Table 10-2

No.	Name
1	Radial trough
2	Dispersion table

10.2.1.2 Radial trough

For interference between the radial troughs, check the circled portion shown in figure below.

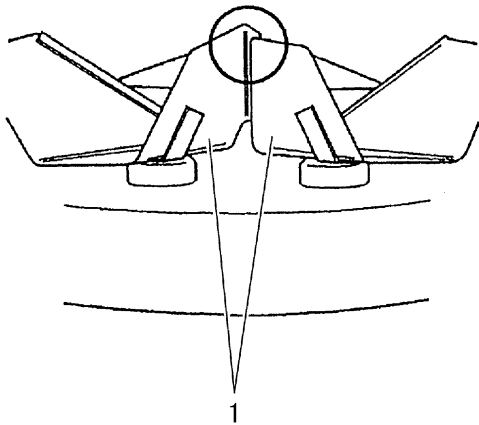


Fig. 10-2

Table 10-3

No.	Name
1	Radial trough

10.2.1.3 Confirming the dew-point checker of the air dryer (WP specification)

Confirm performance of the air dryer by checking the grain color of the dew point checker. When the color is blue, the performance is normal. When the color is either pink, white, or brown, the performance is degraded. Refer to "11.4 Troubleshooting" for detailed handling.

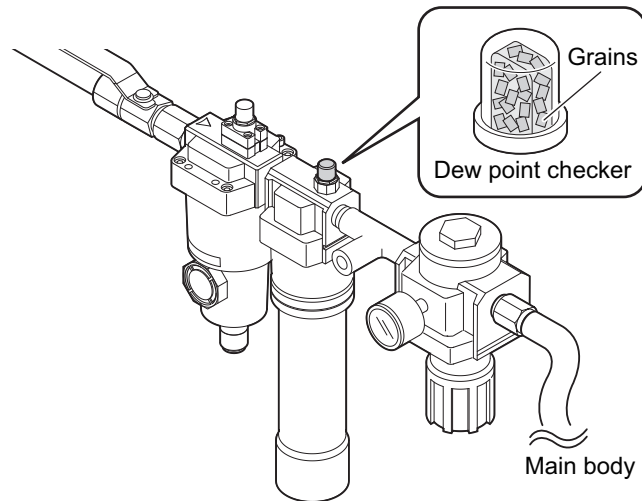


Fig. 10-3

NOTE

- It takes about an hour from starting to supplying the air to the reaction of the grain color.

10.3 Periodical Inspection

Perform maintenance and inspections by following "Table 10-4" in order to maintain the weighing machine in good condition and enhance the product efficiency.

Table 10-4

Inspection Item	Cycle	Reference Section
Inspection of the span	Monthly	"10.3.1 Inspection of Span (Monthly)"
Inspection of the open/close for each hopper gate	Monthly	"10.3.3 Inspection of Open/Close for Hopper Gate (Monthly)"
Inspection of the open/close roller for each hopper	Monthly	"10.3.4 Inspection of Open/Close Roller for Hopper (Monthly)"
Inspection of the amplitude strength for radial feeder	Monthly	"10.3.5 Inspection of Amplitude Strength for Radial Feeder (Monthly)"
Inspection for cracks	Annually	"10.3.6 Inspection for Cracks (Annually)"
Inspection of the rubber cover	Annually	"10.3.7 Inspection of Rubber Cover (Annually)"
Replacement of the memory backup battery	Approx. every five years	"10.3.8 Replacement of Memory Backup Battery"
Replacement of fuses	As needed	"10.3.9 Replacement of Fuses"
Replacement of the air dryer element(WP specification)	Annually	"10.3.10 Replacement of the Air Dryer Element (WP specification)"
Replacement of the air dryer module set(WP specification)	Annually	"10.3.11 Replacement of the Air Dryer Module Set (WP specification)"

10.3.1 Inspection of Span (Monthly)

The Zero Adjustment screen is used to check the span.

NOTE

- Make sure the control unit power switch is ON. If not, press the [Power] key to turn the control unit power switch ON.

- On the Main Menu, press the [Zero Adjst] key.
 - The Zero Adjustment screen appears.
 - All hoppers are selected and displayed in blue.



Fig. 10-4

2. Press the [Start] key.

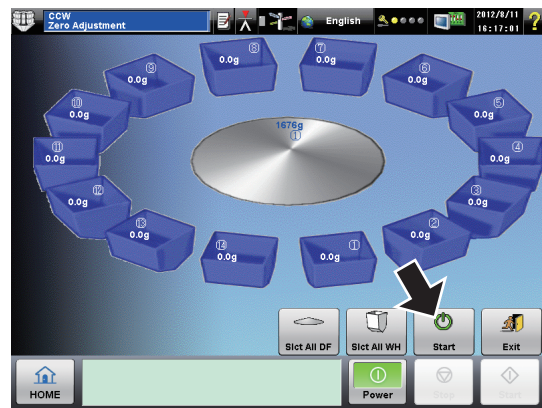


Fig. 10-5

- ▶ The message "Please wait a moment." appears on the screen, and zero adjustment starts.

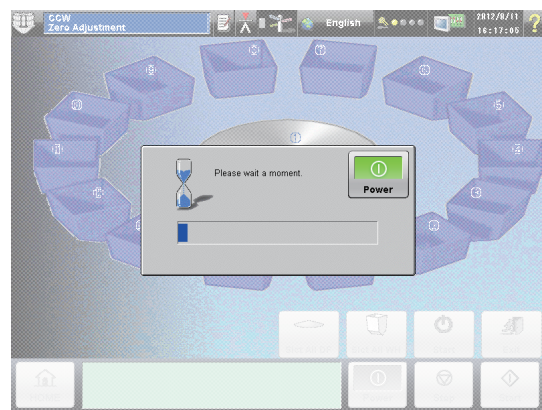


Fig. 10-6

- ▶ When zero adjustment operation finishes, each head displays the respective weight.
3. Make sure that weight display of each head is within 0.0 ± 0.1 g.

NOTE

- If the weight display reading exceeds 0.1 g or falls below -0.1 g, perform the zero adjustment again.
- Pressing each [Head] key on the screen can switch the hopper status to selected or deselected.

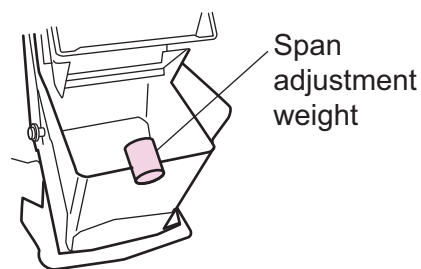


Fig. 10-7

4. Place the span adjustment weight on all of the weigh hoppers.

NOTE

- Make sure that the weight does not make contact with other portions.
- Normally the span adjustment weight is 200 g. However, it may vary depending on the weighing machine specifications.

**Fig. 10-8**

- ▶ The weight of each head is displayed.
- ▶ Make sure that weight display of each head is 200.0 ± 0.1 g.

NOTE

- If the weight display exceeds 200.1 g or falls below 199.9 g, perform the span adjustment (Refer to "10.3.2 Span Adjustment").

5. Remove the span adjustment weight.
6. Press the [Exit] key.
 - ▶ The screen previously displayed appears.
 - ▶ The inspection for span is completed.

**Fig. 10-9**

10.3.2 Span Adjustment

NOTE

- The span adjustment is performed by the "Site Engineer" or higher level personnel.

The span adjustment is performed in order to store the weight with the span adjustment weight (normally 200 g/piece) set to the weigh hopper, into the weighing instrument as 200.0 g. It needs to be adjusted when the weight display shows outside the specified value at the span inspection.

For the span adjustment, follow the procedures below.

1. Check that there are no objects in the weigh hopper.
2. Display the Main Menu for the Site Engineer level or higher.
3. Press the [Machine Set] pop-up key.

► The Machine Set pop-up menu appears.



Fig. 10-10

4. Select "Manual Adjustment".
5. If the [Weighing Adjst] tab is not selected, press the [Weighing Adjst] tab.

► The weight of each head is displayed.



Fig. 10-11

6. To perform the zero adjustment for all weigh hoppers, press [All Head SLCT/CLR] key.

► All [Head] keys are displayed in blue.

TIP

- To perform the zero adjustment for the selected weigh hopper, press the [Head] key of the relevant hopper.

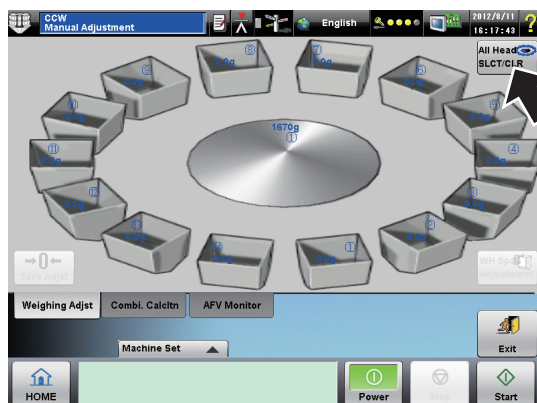


Fig. 10-12

7. Press the [Zero Adjst] key.

► The zero adjustment will begin.

► The display of the zero-adjusted [Head] key becomes normal, and the weight is displayed.

8. Make sure that the weight display is 0.0 ± 0.1 g.

► The zero adjustment is completed.



Fig. 10-13

NOTE

- Make sure that the span adjustment weight does not make contact with other portions.

9. Place the span adjustment weight (normally 200 g/piece) on the weigh hopper that needs the span adjustment.

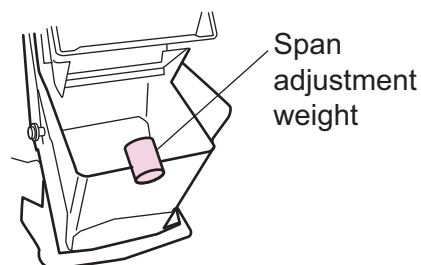


Fig. 10-14

10. To perform the span adjustment for all the weigh hoppers, press the [All Head SLCT/CLR] key.

► All [Head] keys are displayed in blue.

TIP

- To perform the span adjustment for the selected weigh hopper, press the [Head] key of the relevant hopper.

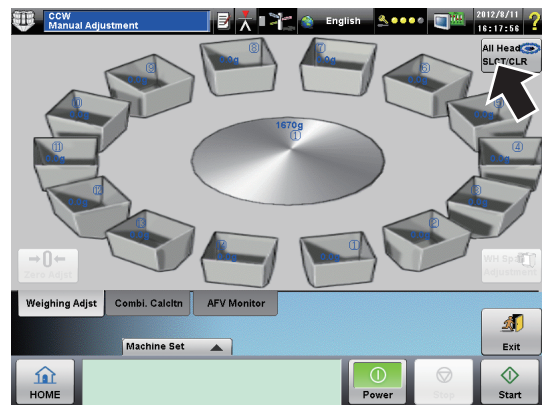


Fig. 10-15

11. Press the [WH Span Adjustment] key.

- The span adjustment starts.
- The display of the span-adjusted [Head] key becomes normal, and the weight is displayed.

12. If the weight display exceeds 200.1 g or falls below 199.9 g, perform the span adjustment, following steps 10 and 11.

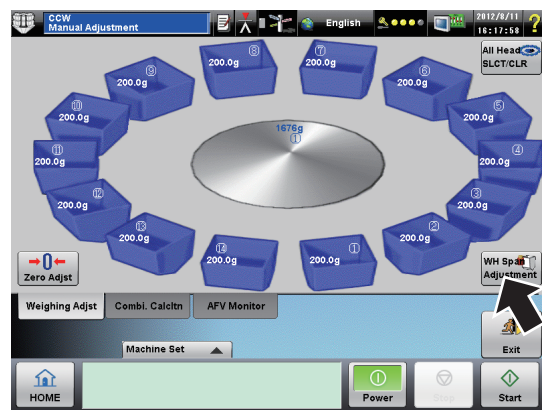


Fig. 10-16

13. When the weight display is within 200.0 ± 0.1 g, remove the span adjustment weight from the weigh hopper.

14. Perform the zero adjustment (steps 6 to 8).

15. Press the [Exit] key.

- The Main Menu appears.
- The span adjustment is completed.



Fig. 10-17

10.3.3 Inspection of Open/Close for Hopper Gate (Monthly)

1. Remove the pool hopper and weigh hopper.

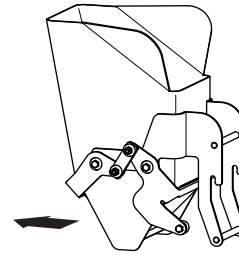


Fig. 10-18

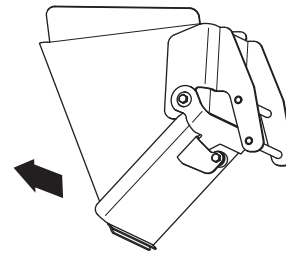


Fig. 10-19

2. Push and pull the open/close lever of the open/close roller for each hopper, and check that each gate opens/closes smoothly.
3. If each gate does not open/close smoothly, check for the distortion of the hopper.
4. Check the wear condition of the bushing used for the supporting point of each gate. If any, replace with new parts.

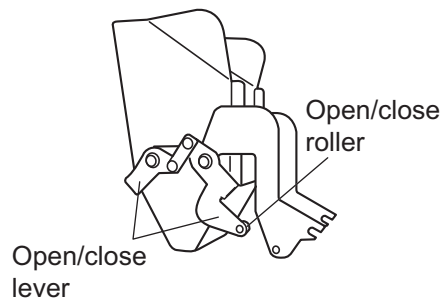


Fig. 10-20

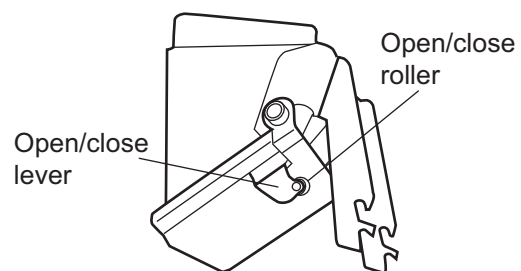


Fig. 10-21

10.3.4 Inspection of Open/Close Roller for Hopper (Monthly)

1. Turn the open/close roller for each hopper by hand, and check that it turns smoothly (Refer to the figure in the previous section).
2. If it does not turn smoothly, check the following items.
Check if the roller has any uneven wear.
Check if the open/close roller has any dirt.
3. If the open/close roller does not turn smoothly even after being cleaned, or if it has uneven wear, replace with new parts.

10.3.5 Inspection of Amplitude Strength for Radial Feeder (Monthly)

NOTE

- If there is a failure in transferring the product on the radial trough, it is possible that the radial trough and trough installation portion have cracks. Check if there are any cracks.
- When the amplitude strength differs, take the amplitude strength value into account when making a judgment.

1. Set the amplitude time to the same value for all the radial feeders using the feeder adjustment function.
2. Start the operation, and check the transfer condition of the products on the radial trough.

NOTE

- Feeders with especially slow transfer speeds need to be adjusted. Contact the distributor or Ishida customer support.

10.3.6 Inspection for Cracks (Annually)

1. Check for cracks on the pool hopper, weigh hopper, and radial trough.
2. Replace the pool hopper, weigh hopper and/or radial trough that have cracks.
3. For possible crack portions, check the circled areas shown in the figure below.

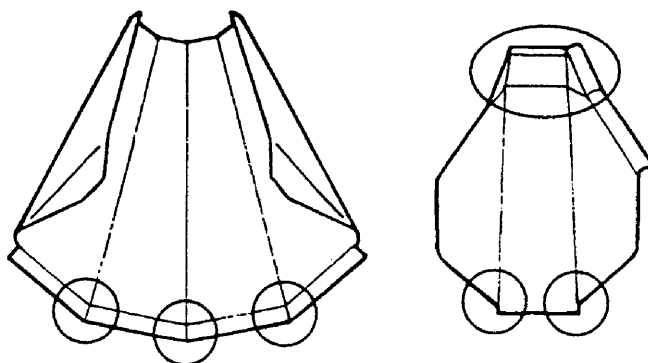


Fig. 10-22

10.3.7 Inspection of Rubber Cover (Annually)

Check for wear, distortion, chap and cracks on the rubber cover of the radial feeder.

NOTE

- The rubber cover with the failure above needs to be replaced.
Contact the distributor or Ishida customer support.

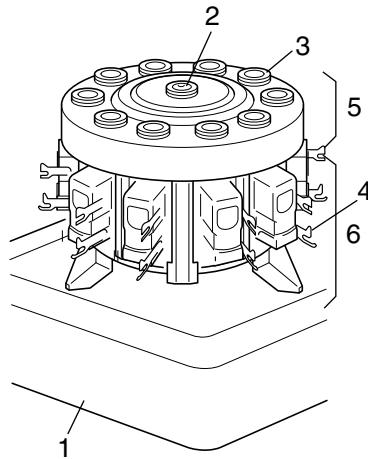


Fig. 10-23

Table 10-5

No.	Name
1	Main body
2	Rubber cover of the dispersion feeder
3	Rubber cover of the radial feeder
4	WH hanger
5	Dispersion portion
6	Drive unit

10.3.8 Replacement of Memory Backup Battery

This weighing machine uses a primary battery as a backup battery in order to store presettings. When the battery voltage decreases due to the battery life, presettings will be deleted.

The weighing machine has a check function for battery voltage, but by the time the battery voltage decrease is detected, presettings may already be deleted.

In order to protect presettings by replacing the battery before the battery voltage decrease due to battery life is detected, it is recommended to periodically contact and ask the distributor or Ishida customer support to replace the battery for your weighing machine.

The rough standard for the battery replacement is every five years. However, the battery life may become less than five years depending on the operation environment.

WARNING

- **The replacement of the memory backup battery is performed by the Ishida service department.
Contact the distributor or Ishida customer support.**

10.3.9 Replacement of Fuses

The following fuses are used for the weighing machine. If a fuse blows out, check the cause, and replace with a new fuse.

Ø5 × 20 250 V 5 A
Ø5 × 20 250 V 3.15 A

DANGER

- **The replacement of the fuses is performed only by the Ishida service department.
Contact the distributor or Ishida Customer Support.**

WARNING

- **Before starting the procedures, always turn OFF the main power switch.**
- **Wait for five minutes after turning the power OFF, and then start the procedures.**

For fuse replacement, follow the procedures below.

- Fuses are used for the AC fuse board (P-5507*) and DC fuse board (P-5508*). Refer to the appendix for the details of each board.
 - Use the tester for the energization check.
1. Remove the fuse board cover.
 2. Turn the fuse holder to the left, and remove the blown-out fuse.

3. Install a new fuse.
4. Put the fuse holder back to its original position.

NOTE

- Turn ON the main power switch at this point. If the LED for the fuse is off, the fuse is normal. If the LED lights up, it means a fuse is still blown out.

5. Reattach the cover.

►The replacement of the fuse is completed.

10.3.10 Replacement of the Air Dryer Element (WP specification)

1. Shut the air supply cockstop and turn the drain outlet to let the air inside the air dryer, then confirm that the pressure gauge reads 0 Mpa.

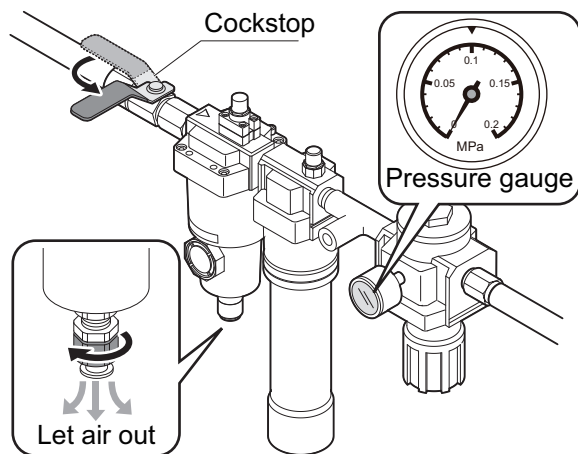


Fig. 10-24

2. Unscrew the four screws to remove the separator.

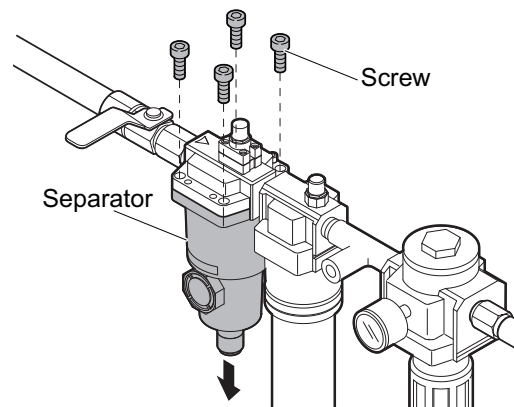


Fig. 10-25

3. Turn the old element counterclockwise to remove from the body. Attach a new element by turning it clockwise.
Re-screw the screws in the reverse order of the step 2.
4. Replacement of the air dryer element is completed.

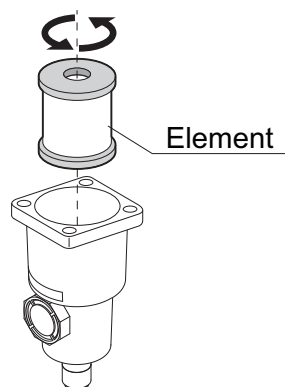


Fig. 10-26

10.3.11 Replacement of the Air Dryer Module Set (WP specification)

1. Shut the air supply cockstop and turn the drain to let the air inside the air dryer, then confirm that the pressure gauge reads 0 Mpa.

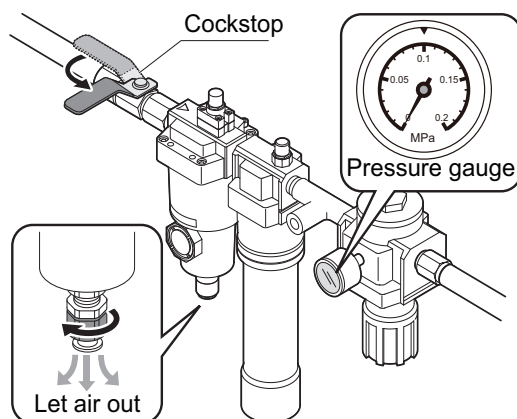


Fig. 10-27

2. Unscrew the four screws to remove the separator.

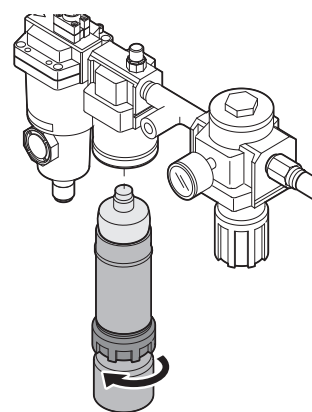


Fig. 10-28

3. Check the packing and spacer are attached, then set the new module to the case.

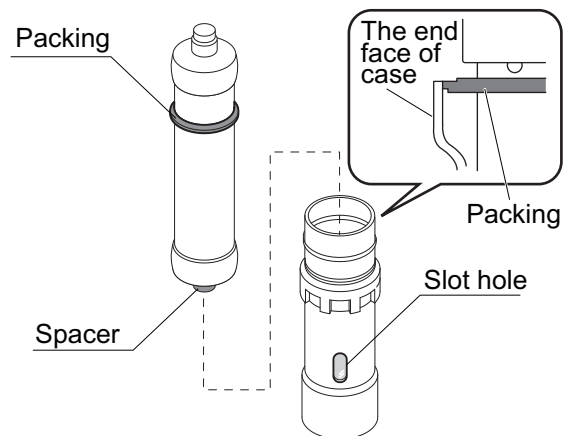
Ensure round hole of module is not seen from slot hole.

NOTE

- Set the packing properly into the end face of case. Do not insert the packing all the way.

4. Re-screw the screws in the reverse order of the step 2.

► Replacement of the air dryer module set is completed.

**Fig. 10-29**